# Literature Review — Domain + ML

**Optical Camera Communication (OCC) Video Message Decoding**

## 1. Domain Fundamentals

Optical Camera Communication (OCC) repurposes a standard camera's image sensor as a communications receiver for a light-based transmitter. Most cameras used in OCC studies are rolling-shutter devices: the sensor exposes rows sequentially rather than all at once, so a fast-blinking LED produces bright/dark stripe patterns within a single frame rather than one on/off value per frame. This is the mechanism the supplied videos rely on for On-Off Keying (OOK) transmission, and it is the reason symbol synchronization, recovering the timing relationship between the transmitter's clock and the camera's row-exposure timing, is treated as a central challenge across the literature reviewed below, alongside robustness to ambient light and motion.

## 2. Comparison Table — Methods, Metrics, Results, Limitations

| Paper (Author, Year) | Approach / Contribution | Metric(s) Used | Reported Result | Limitation Relevant to This Project |
|---|---|---|---|---|
| Liu Y., Chow C.W., Liang K. et al. (2016) — Optics Express | Compares fixed vs. adaptive thresholding schemes for VLC/OCC decoding using a mobile-phone image sensor. | BER, decoding accuracy | Adaptive thresholding consistently outperforms a single fixed threshold under varying ambient light. | Purely threshold-based; no signal processing/ synchronization stage, so degrades under motion or rolling-shutter distortion. |
| Ferrandiz-Lahuerta J., Camps-Mur D., Paradells J. (2015) — IEEE Globecom | Reliable asynchronous VLC protocol exploiting rolling-shutter effect on an off-the-shelf smartphone. | Data rate, operating distance | Achieved up to 700 bps at operating distances up to 3 m using a Nexus 5 camera. | Rate and range are tied to a single tested device; asynchronous framing design is protocol-specific rather than generalpurpose. |
| Kookmin Univ. group (Jang et al., asynchronous OCC studies, 2015–2018) — IEICE/IEEE 802.15 contributions | Asynchronous OCC decoding based on cycle-pattern characteristics when transmitter clock and camera frame rate are mismatched. | BER, symbol rate (ksps) | Error-free transmission (BER < 1e-4) at 4.9–7.5 ksps using smartphone cameras at 30 fps. | Requires the symbol-length-to-lineinterval ratio to be expressible as an integer ratio; sensitive to frame-rate instability. |

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| Liu A.Q., Shi W.X., Ouyang M. et al. (2022) — IEEE J. Lightwave Technology | Comprehensive system model relating LED optical power, distance, and camera exposure to final pixel intensity values. | Pixel-value prediction accuracy vs. distance/exposure | Model quantitatively links transmission distance and exposure time to pixel intensity, guiding threshold selection. | A theoretical/simulation model; validated conditions may not capture all real-world noise sources present in supplied videos. |
| Zhang J., Shi W.X., Wang Q., Liu A., Liu W. (2023) — Optoelectronics Letters | Adaptive-threshold decoding extended with multi-feature LEDstate classification (average gray ratio, gradient radial inwardness). | BER vs. distance | BER reduced from $1\times10^{2}$ (simple $^{-}$ threshold) to $5\times10^{4}$ (multi-feature) at 80 $^{-}$ m range. | Feature design targets LED-array geometry; needs adaptation for a single blinking-LED, message-decoding setup. |
| Jurado-Verdu C., Guerra V., Rabadan J., Perez-Jimenez R. (2022) — Optics Express | Deep-learning estimator for transmitter clock and camera exposure parameters directly from rolling-shutter images. | Relative estimation error | Validated on 7000+ real images; relative errors below 1% for clock/exposure estimation. | Requires training data (synthetic + real) and a DL estimator stage — heavier than the compulsory software-only scope needs as a first pass. |
| Sitanggang O.S., Nguyen V.L., Nguyen H. et al. (2023) — Sensors (2D MIMO C-OOK, deep learning) | Deep-learning-based decoding for MIMO camera-OOK (C-OOK) schemes, building on the IEEE 802.15.7 MIMO C-OOK architecture. | Decoding accuracy under mobility | DL-assisted decoding improves robustness over rule-based C-OOK decoding under receiver/transmitter motion. | Designed for multi-transmitter MIMO arrays; this project has effectively a single-transmitter link, so only the MLdecoding idea transfers, not the MIMO framework. |
| Do T.H., Yoo M. (2021) — IET Optoelectronics | Performance enhancement of OCC via dedicated coding scheme plus region-of-interest (ROI) detection. | BER, ROI-detection accuracy | Combined coding + ROI detection improves decoding performance over ROIagnostic baselines. | Depends on a specific error-correcting code layered on OOK; assumes the code is known, which may not match this project's message format. |
| Cahyadi W.A., Chung Y.H., Ghassemlooy Z. et al. (2020) — Electronics (review) | Survey of OCC principles, modulation schemes (incl. OOK), and open challenges. | N/A (survey) | Consolidates OOK/rolling-shutter fundamentals and identifies synchronization and ambient-light robustness as recurring open problems. | Survey-level; no single reproducible decoding pipeline, but confirms the risks already identified for this project (sync drift, flicker). |

## 3. Candidate Methods for This Project

- Fixed / adaptive threshold baseline — direct application of the thresholding comparison in Liu et al. (2016); fast, interpretable, and the mandatory reference point before any more complex method.
- Signal-processing pipeline — adaptive/local thresholding plus smoothing and explicit symbol-timing recovery, informed by the asynchronous-decoding cycle-pattern approach (Kookmin group) and the pixel-intensity system model (Liu A.Q. et al., 2022) for principled threshold selection.
- Multi-feature decoding refinement — adapting the average-gray-ratio / gradient-based feature idea from Zhang et al. (2023) if a single fixed/adaptive threshold proves insufficient on noisier videos.
- Simple ML-assisted symbol classifier — a lightweight classical classifier (e.g., k-NN, SVM, or Random Forest) on engineered signal features, used only where labelled symbol segments permit, in the spirit of the DL-assisted decoding direction shown by Jurado-Verdu et al. (2022) and Sitanggang et al. (2023) but scaled down to this project's compulsory, dataset-driven scope.

## 4. Key Technical Risks Identified from the Literature

- Synchronization / clock mismatch between transmitter blink rate and camera frame rate — the most consistently reported challenge across the reviewed papers.
- Ambient light and flicker interference, which can mask or mimic the transmitted OOK signal.
- Rolling-shutter exposure and motion effects distorting the stripe pattern used for intensity extraction.
- Limited or absent ground-truth labels restricting how much supervised ML decoding is feasible versus rule-based signal processing.

## 5. Next Step

- With the domain grounded and candidate methods shortlisted, Day 3 moves into research framing for Mentor Review 1: refining the measurable research question, confirming target definition, dataset structure, and evaluation metrics with the mentor before dataset-dependent work begins.